The Autocorrelation Mirage: Dependence-Aware Evaluation for Panel-Expanded Event Forecasting

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Abstract. Trustworthy event forecasting in panel-expanded data depends on evaluation protocols that respect the dependence structure of episodes. A common workaround for small numbers of independent episodes expands each episode into daily observations, but this changes the evaluation problem: row-wise cross-validation can place observations from the same episode in both training and test sets, allowing a flexible model to exploit episode-specific dependence in settings where the intended target is generalization to unseen episodes. We call this mechanism *episode memorization.* We distinguish it from two related problems that are often conflated with it: explicit future-information leakage in features and preprocessing leakage from operations such as normalization or scaling. In a controlled benchmark with strictly causal features and a boosted-tree classifier, row-wise five-fold cross-validation increases mean AUC by 0.28, from 0.55 to 0.84. Under episode-grouped evaluation, an explicit time-to-event leak increases AUC by 0.26, from 0.55 to 0.81. A compact robustness grid over three episode counts, three dependence levels, and three feature sizes shows that the mean Δ_{CV} gap remains near zero for logistic regression but is substantially larger for random forests and boosted trees, especially once dependence is moderate or strong. Although the title emphasizes autocorrelation, the mechanism is broader: latent episode heterogeneity and shared event-time structure can also create split sensitivity when serial autocorrelation is weak. Under a second DGP with monotone pre-event drift, full-trajectory episode normalization inflates grouped-CV AUC for logistic regression from 0.445 to 0.641. We therefore propose Δ_{CV}, the gap between row-wise and episode-grouped estimates, as a practical warning signal rather than a universal threshold, and we provide a dependence-aware evaluation protocol based on episode-grouped splitting and fold-local preprocessing. The contribution is methodological: a compact taxonomy, a mechanistic explanation, and a low-cost protocol for more trustworthy panel-expanded event forecasting.

Keywords: pseudoreplication, cross-validation, panel data, event forecasting, episode memorization

1 Introduction

Trustworthy AI systems depend not only on model capacity, but also on evaluation protocols that match the deployment target. This is especially important in panel-expanded event forecasting, where the number of independent episodes is small but decisions may still rely on reported predictive accuracy. In domains such as conflict forecasting [5, 1], clinical deterioration [8], and predictive maintenance, analysts may observe only 20–50 independent episodes with 10–20 features and then expand each episode into daily rows, turning a small set of evaluation units into hundreds of observations. That is already a small-data regime by conventional events-per-variable heuristics for generalized linear models [7, 10].

That expansion does not simply increase sample size; it changes the evaluation problem. When row-wise K-Fold cross-validation assigns expanded observations to folds independently, rows from the same episode can appear in both training and test sets. Because consecutive observations $X_{e,t}$ and $X_{e,t+1}$ are autocorrelated, a flexible model can recognize episode-specific trajectory structure rather than infer risk that transfers to unseen episodes. We call this mechanism **episode memorization**. In that setting, row-wise CV turns a forecasting task (generalizing to new episodes) into a partially interpolative task (matching rows within known episodes).

It is useful to distinguish three failure modes that are often conflated. *Explicit leakage* occurs when a feature directly encodes unavailable future information such as $T_e - t$. *Normalization leakage* occurs when test-fold statistics or future within-episode summaries enter preprocessing even though they would be unavailable at prediction time. *Pseudoreplication* occurs when dependent rows from the same episode are split across training and test folds. The first two are feature- or preprocessing-level violations. The third is an evaluation-design violation and can arise even when every feature is strictly causal.

In our simulation, replacing grouped five-fold evaluation with row-wise `KFold` increases mean AUC by 0.283, from 0.552 to 0.835. In the same benchmark, an explicit time-to-event leak increases grouped-evaluation AUC by 0.261, from 0.552 to 0.813. We present this comparison as evidence that pseudoreplication can be severe, not as a universal ranking across all domains or models. Kaufman et al. [4] formalized data leakage and learn-predict separation in i.i.d. settings. Roberts et al. [9] argued for block cross-validation for structured data. Hurlbert [2] coined *pseudoreplication* in experimental design. Kapoor and Narayanan [3] documented leakage across many applied ML papers, and Neunhoeffer and Sternberg [6] showed that cross-validation misuse can overturn substantive forecasting claims. We build on these strands in the specific setting of panel-expanded event forecasting, where dependence-aware evaluation is both necessary and easy to operationalize.

The paper makes five contributions. First, it clarifies the distinction between explicit leakage, preprocessing leakage, and pseudoreplication in expanded panels. Second, it formalizes episode memorization as the mechanism by which row-wise CV can produce optimistic estimates. Third, it quantifies the resulting optimism in a controlled benchmark with strictly causal features. Fourth, it stress-tests that optimism over a compact grid of episode counts, dependence levels, dimensionalities, and model families. Fifth, it proposes Δ_{CV}, the gap between row-wise and grouped CV estimates, as a practical warning signal together with a dependence-aware evaluation protocol.

The goal is methodological rather than epidemiological: we identify a failure mode, explain its mechanism, and quantify its potential impact under controlled conditions. We do not attempt to measure how prevalent this problem is in published practice; instead, we conclude with a compact audit checklist that authors and reviewers can apply to panel-expanded event forecasting studies.

2 Theoretical Framework

Consider E episodes indexed by e. Each has time $t \in \{0, \dots, T_{\max}\}$, administrative censoring time C_e, observed-event indicator $\delta_e := \mathbf{1}\{\text{event observed by } C_e\}$, features $X_{e,t} \in \mathbb{R}^p$, and static attributes α_e. When $\delta_e = 1$, let T_e denote the observed event time. For horizon H, eligible rows are

$$\mathcal{T}_e(H) := \{t : \delta_e = 1,\ t < T_e\} \cup \{t : \delta_e = 0,\ t + H \leq C_e\}.$$

The binary target is

$$Y_{e,t}(H) := \begin{cases} \mathbf{1}\{T_e \leq t + H\}, & \delta_e = 1, \\ 0, & \delta_e = 0 \text{ and } t + H \leq C_e. \end{cases}$$

Late censored rows with $t+H > C_e$ are excluded from the supervised evaluation set rather than labeled heuristically.

Let $\mathcal{F}_{e,t^-} := \sigma(\{X_{e,s} : s < t\} \cup \{\alpha_e\} \cup \mathcal{E}_{<t})$ denote information strictly before t.

Definition 1 (Predictable Feature). *Feature $X_{e,t}$ is* predictable *if it is $\mathcal{F}_{e,t^-}$-measurable, that is, determined by information available before t.*

This extends Kaufman et al.'s [4] learn-predict separation to filtrations. Violations need not be explicit: global statistics computed over $\{X_{e,s} : s \leq T_e\}$ implicitly depend on T_e.

Definition 2 (Explicit Leakage). *An observation exhibits* explicit leakage *if a feature is not predictable because it is a direct function of future outcome information such as $T_e - t$.*

Definition 3 (Normalization Leakage). Normalization leakage *occurs when test-fold statistics, future within-episode values, or full-trajectory episode summaries enter transformed inputs even though they would be unavailable at prediction time.*

Definition 4 (Pseudoreplication). Pseudoreplication *occurs when the evaluation split places observations from the same episode in both training and test folds, violating the independence assumptions underlying row-wise validation even if all features are predictable.*

The deployment target for unseen episodes is the episode-level risk

$$R_{\text{episode}}(f) = \mathbb{E}_{e^\star}\left[\frac{1}{|\mathcal{T}_{e^\star}|}\sum_{t\in\mathcal{T}_{e^\star}} \ell\{f(X_{e^\star,t}), Y_{e^\star,t}\}\right],$$

where $e^\star$ is not represented in the training episodes. Episode-grouped validation estimates this target by holding out complete episodes. Row-wise validation instead estimates a conditional interpolation risk closer to

$$R_{\text{row}}(f) = \mathbb{E}\left[\ell\{f(X_{e,t}), Y_{e,t}\} \mid e \in \mathcal{E}_{\text{train}}\right],$$

because test rows may share latent episode effects, trajectory structure, and event time with training rows. Thus the issue is not dependence alone; it is a mismatch between the generalization unit implied by deployment and the unit used by validation.

Theorem 1 (Explicit leakage can make ranking nearly perfect). *Suppose both classes have positive probability and the leaked score has the form $S_{e,t} = f(T_e - t) + \epsilon_{e,t}$, where f is strictly monotone on the support relevant for $Y_{e,t}(H)$ and the learned scoring rule can use the sign convention that smaller time-to-event implies higher risk. If $\mathrm{Var}(\epsilon_{e,t}) = \sigma^2$, then as $\sigma^2 \to 0$ the leaked score perfectly ranks positive horizon rows above negative rows, so* AUC $\to 1$.

For this theorem we restrict attention to eligible rows from episodes with observed event times; equivalently, censored no-event rows can be treated as having $T_e - t = \infty$ when included.

Proof sketch. Since $Y_{e,t}(H) = 1$ iff $T_e - t \in (0, H]$, a strictly monotone transform of $T_e - t$ induces the same ordering. As the noise variance vanishes, the leaked score preserves that ordering almost surely and therefore achieves perfect ranking. □

3 Implicit Violations in Standard Pipelines

Table 1 summarizes the three failure modes emphasized in this paper. Explicit leakage and normalization leakage are violations of predictability. Pseudoreplication is different: it is an evaluation mismatch induced by splitting dependent rows across folds.

Normalization traps are heterogeneous. Simple global standardization $\tilde{X} = (X - \mu)/\sigma$ can be weak when the feature process is approximately stationary, but transforms that use future within-episode values—for example, full-trajectory episode means or episode-wise z-scores—can be much more problematic. Mean imputation uses future observations; episode-mean imputation uses $\bar{X}_e$, which also depends on T_e. Centered moving averages use future values by construction. These traps require features with pre-existing trends toward the event. Strictly causal stationary features are often less sensitive, but many real-world variables are not: vital signs may deteriorate before death, equipment diagnostics may worsen before failure, and conflict indicators may escalate before violence.

Table 1. Three distinct sources of evaluation optimism in expanded panel forecasting.

Failure mode	**Mechanism**	**Future info?**	**Primary fix**
Explicit leakage	Feature directly reveals a post-t outcome signal such as $T_e - t$	Yes	Redesign or remove feature
Normalization leakage	Test-fold statistics, future within-episode values, or full-trajectory episode summaries enter transformed inputs	Indirectly	Fit transforms within each training fold; for online use, rely on trailing or historically available statistics
Pseudoreplication	Rows from the same episode appear in both train and test folds	No	Group by episode at split time

4 Episode Memorization Under Row-Wise Cross-Validation

Even with leak-free features, standard K-Fold CV can inflate performance through *episode memorization* when the target is generalization to unseen episodes. This is Hurlbert's [2] pseudoreplication translated to ML: "replicates are not statistically independent." Within-episode observations share: (1) latent episode effects α_e; (2) autocorrelated features with $\mathrm{Corr}(X_{e,t}, X_{e,t+1}) \gg 0$; (3) labels determined by the same T_e. When row-wise CV places observations from episode e in both training and test sets, the model learns episode-specific patterns—the "texture" of each episode's autocorrelation structure. At test time, it performs *identity inference*: recognizing which episode an observation belongs to, then recalling the associated outcome. This transforms forecasting (extrapolation) into interpolation (pattern-matching). As a descriptive design-effect approximation, if each episode contributes approximately m rows and the average within-episode pairwise correlation is $\bar{\rho}$, then

$$n_{\text{eff}} \approx \frac{n}{1 + (m-1)\bar{\rho}}.$$

This approximation is not used for inference; it is reported only to highlight that panel expansion can substantially overstate the amount of independent evidence.

A minimal example makes the mechanism concrete. Suppose episodes A, B, and C each contribute two daily rows. A random two-fold split may train on $\{A_1, B_1, C_2\}$ and test on $\{A_2, B_2, C_1\}$; every test row now has a sibling row from the same episode in training. A grouped split instead trains on $\{A_1, A_2, B_1, B_2\}$ and tests on $\{C_1, C_2\}$, forcing prediction to transfer across episodes rather than match within them.

For $n = 600$, $m \approx 20$, and $\bar{\rho} \approx 0.5$, this approximation gives $n_{\text{eff}} \approx 55$—a 10× reduction. But the inflation from episode memorization is *not* merely a sample-size effect: in the empirical benchmark below, row-wise CV still achieves AUC above 0.8 despite the small effective sample size.

This suggests an empirical expectation that, once within-episode dependence is present, row-wise CV will exceed grouped CV more strongly for higher-capacity models. Section 5 tests that expectation directly over a compact grid of dependence levels, episode counts, dimensionalities, and model families.

Proposition 1 (Sibling overlap under shuffled row-wise K-fold). *Consider an episode with m_e eligible rows assigned independently to one of K folds. Condition on one row being placed in the test fold. Then the probability that at least one of its $m_e - 1$ sibling rows appears in the training folds is*

$$\Pr(\textit{sibling overlap}) = 1 - K^{-(m_e-1)},$$

and the expected number of same-episode training siblings is

$$\mathbb{E}[N_{\textit{sibling in train}}] = (m_e - 1)\frac{K-1}{K}.$$

Proof sketch. Condition on the test row being assigned to fold k. Its siblings avoid the training set only if they also fall in fold k, which under independent fold assignment occurs with probability $K^{-(m_e-1)}$. The complement gives the overlap probability, and linearity of expectation yields the expected sibling count. The exact probability differs slightly for deterministic balanced K-Fold, but the approximation captures the usual shuffled row-wise setting. □

5 Empirical Validation

We generate $E = 30$ episodes with: episode effect $\alpha_e \sim \mathcal{N}(0, 0.5)$; AR(1) latent process $Z_{e,t} = 0.7Z_{e,t-1} + \epsilon_t$; hazard $h_{e,t} = \text{logit}^{-1}(-3 + \alpha_e + 0.15Z_{e,t})$; features from $Z_{e,<t}$ only (strictly causal); and administrative censoring $C_e = T_{\max} = 60$. For observed-event episodes ($\delta_e = 1$), eligible rows satisfy $t < T_e$ and receive label $Y_{e,t}(14) = \mathbf{1}\{T_e \le t+14\}$. For censored no-event episodes ($\delta_e = 0$), only rows with $t + 14 \le C_e$ are eligible, and those rows receive $Y_{e,t}(14) = 0$; late censored rows are excluded rather than encoded through a synthetic $T_e = C_e + 1$. The intercept -3 corresponds to a low instantaneous daily hazard ($\text{logit}^{-1}(-3) \approx 0.047$), but the 14-day horizon makes the induced row-level label much denser: over 10 seeds, the filtered at-risk panel contains about 587 observations, mean episode length 19.6, episode-event frequency 0.91, and positive-row prevalence 0.48. We therefore frame the paper as panel-expanded event forecasting rather than as a benchmark for ultra-rare row labels. The five base features are strictly causal transformations of $Z_{e,<t}$; for the robustness grid we add independent nuisance features to reach $p \in \{20, 100\}$. The main benchmark uses a histogram gradient boosting classifier with 100 boosting iterations, maximum depth 4, and learning rate 0.1. Grouped and row-wise protocols both use $K = 5$ for the main benchmark: `GroupKFold(n_splits=5)` and shuffled `KFold(n_splits=5)`. The repository also implements a temporal-blocked within-episode splitter for

Table 2. Simulation results over 30 paired replicates. Also shown is a grouped constant-prevalence baseline; the temporal-blocked within-episode baseline is reported in text because it targets a different estimand.

Condition	AUC	Brier	ΔAUC vs. baseline	95% paired CI
Leak-Free + Grouped CV	0.552 ± 0.101	0.330 ± 0.048	baseline	–
No-Skill Prev. + Grouped CV	0.430 ± 0.025	0.245 ± 0.009	-0.122	[-0.159, -0.086]
Leak-Free + Row-wise KFold	**0.835 ± 0.040**	0.162 ± 0.023	**+0.283**	[0.246, 0.320]
Global Norm. + Grouped CV	0.552 ± 0.101	0.330 ± 0.048	+0.000[†]	[0.000, 0.000]
Explicit Leak + Grouped CV	0.813 ± 0.042	0.198 ± 0.041	+0.261	[0.236, 0.286]

[†]See text for interpretation.

the alternative target of forecasting later rows of already observed episodes; we report that baseline below in text because it estimates a different target than unseen-episode generalization. The compact robustness grid below uses grouped and row-wise three-fold CV for tractability. For each replicate, we compute out-of-fold predictions for all eligible rows and then compute a single pooled AUROC and Brier score. To keep longer episodes from dominating diagnostics, the reproducibility package also reports episode-weighted AUROC and episode-mean Brier summaries. This pooled evaluation avoids dropping grouped folds with one class, which can occur under episode-aware validation. Standardization is fit within each training fold except in the deliberate preprocessing-leakage conditions. Because each random seed is reused across grouped, row-wise, normalization, explicit-leak, and no-skill conditions, the reported AUC gaps are paired within replicate. The reported 95% paired intervals use the matched replicate-level differences, $\bar{d} \pm t_{0.975,n-1} s_d/\sqrt{n}$. Because rows within episodes are dependent, we also report episode-bootstrap uncertainty for Δ_{CV} by resampling episode IDs rather than rows. All conditions are paired by random seed, and the same simulated episodes are used across grouped, no-skill, row-wise, normalization, and explicit-leak settings within each replicate. An anonymized reproducibility package accompanies the submission and contains the simulation code, result CSVs, and figure-generation scripts used to regenerate the reported tables and plots. The benchmark table below uses 30 replicates; the compact grid uses two replicates per $E \times \rho \times p \times$ model cell and should therefore be read as an exploratory stress-test of qualitative behavior rather than a claim of asymptotic precision.

Table 2 and Figure 1 reveal the central finding: row-wise cross-validation alone can create substantial optimism. Row-wise KFold yields AUC=0.835 from strictly causal features, an absolute gap of 0.283 over grouped CV (AUC=0.552), with a paired 95% interval of [0.246, 0.320] across the 30 matched replicates. The episode-bootstrap intervals are also strictly positive in all 30 replicates; their mean replicate-wise 95% interval is [0.141, 0.413] (Table 3). In this benchmark, explicit $T_e - t$ leakage raises grouped-evaluation AUC by 0.261 with paired 95% interval [0.236, 0.286]. The grouped constant-prevalence baseline attains

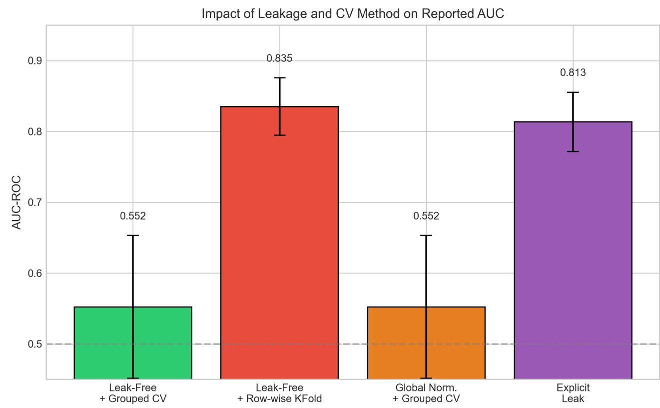

Fig. 1. Main benchmark (30 replicates) under matched five-fold protocols. Row-wise KFold increases AUROC and improves Brier score relative to episode-grouped five-fold validation even though the features are strictly causal.

Table 3. Δ_{CV} uncertainty in the 30-replicate main benchmark. The episode-bootstrap row averages replicate-wise interval endpoints after resampling episodes within replicate.

Summary	Δ_{CV}	95% interval
Matched replicate mean	0.283	[0.246, 0.320]
Episode bootstrap	0.283	[0.141, 0.413]

AUC=0.430 and Brier=0.245. Relative to that baseline, grouped boosting improves ranking but worsens Brier (0.330), indicating weak out-of-sample calibration in this small-E regime even when discrimination improves. We therefore treat Brier as a complementary calibration-sensitive diagnostic rather than the primary target of the pseudoreplication argument. For the alternative target of forecasting later rows within already observed episodes, temporal-blocked within-episode validation yields AUC=0.917 and Brier=0.232; we report it separately because it preserves episode identity and therefore estimates a different risk than unseen-episode grouped CV. Simple global normalization shows no measurable AUC gap because the base DGP is approximately stationary. That row should therefore be read as a conservative stationary benchmark, not as evidence that preprocessing leakage is unimportant. For tree-based learners, simple affine rescaling is also expected to have limited effect; the normalization row is included as a conservative stationary baseline, whereas the drift-DGP experiment illustrates a transform that genuinely uses future within-episode information. The same benchmark still averages only about 68 effective observations despite roughly 587 at-risk rows, underscoring that row expansion does not create independent evidence.

Table 4. Exploratory compact robustness grid: mean Δ_{CV} averaged over $E \in \{20, 50, 100\}$ and $p \in \{5, 20, 100\}$ (2 replicates per cell).

Model family	$\rho = 0.0$	$\rho = 0.6$	$\rho = 0.9$
Logistic regression	0.03	0.03	0.04
Random forest	0.13	0.15	0.16
Boosted trees	0.17	0.23	0.21

Table 4 tests whether the main result is tied to one model or one dimensionality setting. Across 81 $E \times \rho \times p \times$ model cells, higher-capacity tree ensembles exhibit much larger row-wise optimism than logistic regression. Averaged over E and p, mean Δ_{CV} rises from 0.13 to 0.16 for random forests, whereas boosted trees remain between 0.17 and 0.23 and are largest at moderate-to-high dependence. Logistic regression stays near zero by comparison (0.03–0.04). Varying p up to 100 nuisance features does not remove the effect, so the optimism is not confined to low-dimensional settings. Because this grid uses only two replicates per cell, we interpret it as an exploratory stress-test rather than a precise estimate of effect size. The largest individual gaps in the grid are 0.38–0.52 AUC points and occur in the smallest-E, higher-dependence, tree-based conditions.

The nonzero gap at $\rho = 0$ shows that the effect is not driven by autocorrelation alone. Episode-level latent heterogeneity and shared event-time structure can already create row-wise optimism; increasing autocorrelation strengthens the same evaluation-unit mismatch.

Normalization leakage required a second DGP with monotone pre-event drift. In that setting, simple global standardization again had negligible effect, but full-trajectory episode-wise normalization leaked future within-episode values and raised grouped-CV AUC for logistic regression from 0.445 $\pm$ 0.147 to 0.641 $\pm$ 0.014. The much smaller replicate variability in the leaky condition indicates that this shift is stable across seeds in this DGP rather than driven by one anomalous run. Tree ensembles were less sensitive in the same experiment. The practical implication is that preprocessing leakage is real but model-dependent: some transforms are benign in a given benchmark, whereas others can materially inflate grouped evaluation.

Figure 2 complements Table 4 by making the two supporting patterns visually explicit: larger Δ_{CV} values are concentrated in tree-based learners once dependence is moderate or high, and in the drift DGP the largest grouped-CV shift appears for logistic regression under full-trajectory episode normalization.

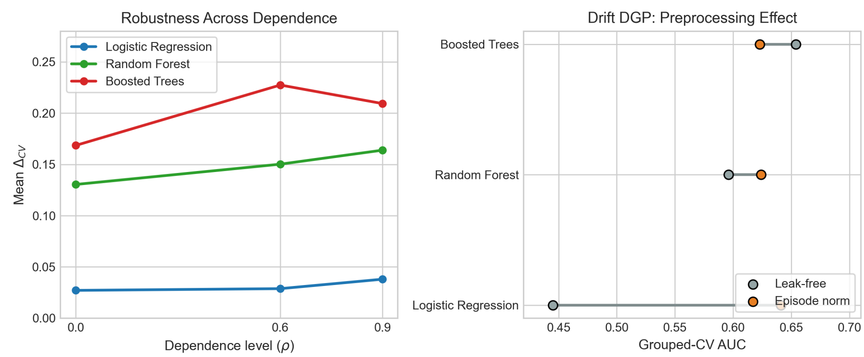

Fig. 2. Companion diagnostics for the strengthening experiments. Left: mean Δ_{CV} versus dependence after averaging over E and p for each model family. Right: grouped-CV AUC in the drift DGP under leak-free inputs and full-trajectory episode normalization; global standardization overlaps the leak-free baseline and is omitted for clarity.

6 The Δ_{CV} Diagnostic and Dependence-Aware Evaluation Protocol

We propose a simple diagnostic for split sensitivity under panel expansion:

Definition 5 (Δ_{CV} diagnostic). *Let $\widehat{\mathrm{AUC}}_{\mathrm{row}}$ denote the pooled out-of-fold AUROC under row-wise KFold and let $\widehat{\mathrm{AUC}}_{\mathrm{group}}$ denote the pooled out-of-fold AUROC under episode-grouped KFold. We define*

$$\Delta_{\mathrm{CV}} = \widehat{\mathrm{AUC}}_{\mathrm{row}} - \widehat{\mathrm{AUC}}_{\mathrm{group}}.$$

Large positive values indicate split sensitivity: the model performs better when test rows are allowed to share episodes with training rows than when entire episodes are held out.

In our main benchmark, $\Delta_{\mathrm{CV}} = 0.283$. In the robustness grid, logistic regression usually stays near zero, whereas random forests and boosted trees remain well above 0.10 on average and grow with ρ. That pattern makes values around 0.05 a useful warning heuristic in our setting, but we do not claim a universal decision threshold. The relevant magnitude will depend on the domain, target definition, dependence strength, and model family. This metric requires no additional data collection—only re-running evaluation with episode-grouped validation instead of row-wise `KFold`.

Dependence-Aware Evaluation Protocol: (1) *Preprocessing:* fit transformations on training folds only; use forward-fill imputation and trailing windows. (2) *Evaluation:* use grouped splitting with `groups=episode_id`; never split episodes across folds. (3) *Reporting:* report Δ_{CV} and n_{eff} alongside AUC.

Minimal algorithm: (1) construct labels only for rows whose H-step outcome is observed; (2) choose the split protocol that matches the deployment

Table 5. Audit checklist for panel-expanded event forecasting studies.

Question	Why it matters
What is the independent episode unit?	Defines the generalization target and the appropriate grouping variable.
Are rows expanded from episodes?	Determines whether row-wise splitting can create dependence overlap.
Can rows from the same episode appear in both train and test folds?	Directly tests for pseudoreplication.
Are preprocessing parameters fit within each training fold?	Detects normalization, imputation, or filtering leakage.
Are both row-wise and grouped estimates reported?	Makes Δ_{CV} observable rather than implicit.

target; (3) fit preprocessing on each training fold only; (4) fit the model and predict held-out rows; (5) aggregate pooled and episode-weighted metrics from aligned out-of-fold predictions; and (6) optionally rerun row-wise KFold and bootstrap episodes to quantify Δ_{CV} uncertainty.

Specificity and failure modes: a large positive Δ_{CV} is a warning flag, not a proof of episode memorization. Distribution shift between grouped folds, severe fold imbalance, episode heterogeneity, or a mismatch between row-wise interpolation and the true deployment target can also enlarge the gap. Near-zero gaps can likewise be false negatives when dependence is weak or model capacity is limited, as illustrated by the logistic-regression cells in Table 4. Practical warning signs are therefore: large positive Δ_{CV}; substantial disagreement between row-wise and grouped CV; and unusually strong discrimination despite a very small number of episodes.

7 Discussion

The main lesson is not that pseudoreplication is always larger than every other leakage mechanism. Rather, row-wise CV can by itself create severe optimism when expanded rows retain strong episode-specific dependence. This matters because row-wise splitters are convenient defaults in general-purpose ML libraries and can be applied mechanically to expanded panels even when they are inappropriate for the underlying prediction target.

Row-wise validation is not always invalid. If the deployment target is interpolation within already observed episodes—for example, forecasting future rows for machines, patients, or regions that are already represented in the training history—then a row-wise or temporally blocked within-episode split may estimate a relevant risk. Our critique applies when the target is generalization to new episodes, where the independent sampling unit is the episode rather than the expanded row.

Table 5 turns the paper's warning into a practical review tool. We do not claim to measure prevalence in the literature here; instead, the checklist highlights the concrete questions that authors, reviewers, and practitioners can apply when auditing panel-expanded forecasting studies.

Our study has limitations. The evidence remains simulation-based and does not establish how prevalent the problem is in published practice. Although the robustness grid now spans linear, bagged-tree, and boosted-tree models, we do not yet include real-data reanalyses or sequence models. A compact public reanalysis or semi-synthetic case study would be the next natural extension once page budget permits. The strongest normalization example used full-trajectory episode-wise normalization on a drift DGP; simple global standardization alone was much weaker in this synthetic setting. Because the label asks whether an event occurs within the next 14 days, the simulated row-level positive prevalence is not ultra-rare even though the underlying daily hazard is low. The paper is therefore best read as a contribution on dependence-aware evaluation for panel-expanded event forecasting rather than as a full study of extreme class imbalance or prevalence in published practice.

Even with these limitations, the implication for trustworthy evaluation is straightforward. Panel expansion creates value only if validation respects episode boundaries. In our benchmark, replacing grouped evaluation with row-wise KFold increases AUC by 0.283 despite strictly causal features, and the compact grid shows that the same qualitative problem persists across episode counts, dimensionalities, and model families, especially for tree-based learners. Reporting grouped CV, fold-local preprocessing, Δ_{CV}, and approximate n_{eff} would make such optimism easier to detect and easier to discuss transparently in future panel-expanded event forecasting studies.

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